

TOC (CFG + PDA) – Numerical Practice Set (VIT Level)

Q1. Construct a CFG for $L = \{ a^i b^j c^k \mid i = k, j \geq 1 \}$. Also give leftmost derivation for aabbbcc and draw parse tree.

Q2. Convert the CFG into CNF: $S \rightarrow ASA \mid aB \mid \epsilon$, $A \rightarrow B \mid S$, $B \rightarrow b \mid \epsilon$.

Q3. Design a PDA for $L = \{ a^n b^m \mid n \geq m \geq 1 \}$ using empty stack acceptance.

Q4. Convert CFG to PDA: $S \rightarrow aSb \mid SS \mid \epsilon$. Explain stack for string aabb.

Q5. Using Pumping Lemma for CFLs, prove $L = \{ a^n b^n c^{(n+1)} \mid n \geq 1 \}$ is not context-free.

Q6. Construct CFG for $L = \{ w \in \{0,1\}^* \mid \#0 = \#1 \}$. Is it ambiguous? Justify.

Q7. Design PDA for palindromes over $\{a,b\}$. Is it deterministic?